What is an Operating System?

- Resource allocator: manages CPU, memory, I/O devices

- Control program: manages execution of user programs

- Kernel: the one program running at all times

Process States:

- Ready: waiting to be assigned to CPU

- Running: instructions are being executed

- Waiting: waiting for an I/O event to occur

- Scheduling algorithms: FCFS, Round Robin, Priority

Virtual Memory Concepts:

- Logical address space mapped to physical address space

- Pages and frames: fixed-size memory blocks

- Page Table: translates logical page numbers to physical frames

Coffman Conditions for Deadlock:

1. Mutual Exclusion

2. Hold and Wait

3. No Preemption

4. Circular Wait

Banker's Algorithm ensures deadlock avoidance.